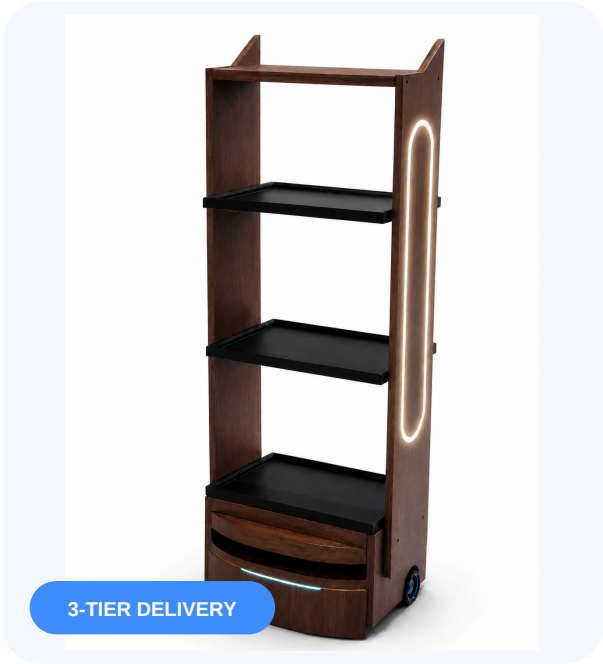


WORKING PROTOTYPE

Restaurant service, made predictable.

A guided autonomous robot that carries meals between the kitchen and tables, follows repeatable routes, verifies its motion, and returns to its dock.

Built for restaurants that need a practical first step into automation - without depending on complex free-space navigation.



- WEB DISPATCH
- MAGNETIC GUIDANCE
- ENCODER + IMU
- AUTO RETURN

A reliable path from kitchen to table

THE OPERATIONAL PROBLEM

Repetitive delivery pulls staff away from guests.

Reliability determines whether automation adds value.

OUR APPROACH

Constrain the route, verify movement, and recover within limits.

Designed for repeatability and easier deployment.

01 **Sense the route**

Seven Hall sensors detect the magnetic line and route markers for centering, junction recognition, and stopping.

02 **Verify motion**

Wheel encoders check segment distance while the MPU6050 controls repeatable 90 and 180 degree turns.

03 **Recover safely**

Line-loss recovery, bounded search windows, motor stop logic, and dock confirmation handle common faults.

PROTOTYPE STATUS

Core navigation is running on hardware

PID line following | Table-2 round trip | IMU turns
Encoder distance checks | Dock and charger detection

NEXT MILESTONES

1 Engineering validation
2 Restaurant pilot
3 Production-ready electronics

THE ASK Seeking a restaurant pilot partner and support for production electronics, safety validation, and manufacturing.

